



Varendra University
Department of Computer Science and Engineering

Undergraduate Project Proposal

Title of the Project

NeuroCommerce: AI-Driven Retail Automation and Inventory Intelligence Platform

Submitted by:

Name: Oloc Chandro (Partha)

ID: 232311131

Partho Kumar Ray

ID: 232311136

Pritom kumar Pramanik

ID: 232311137

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Supervisor:

Barisha Chowdhury

Lecturer

Department of CSE

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1. Introduction

The digital retail industry is rapidly growing; however, let's face the facts—tracking inventory has become a real pain. Many companies are still employing basic spreadsheet management or manual counting. This consumes lots of time, leads to numerous errors, and makes you react rather than act. If your prediction fails, you end up with a warehouse full of products you cannot sell, or vice versa—you have no products to sell and miss your chance to earn a fortune.

Here enters Machine Learning—rather than making predictions by yourself, the system analyzes your historical data and makes forecasts for you. What is more, once the software identifies your clients' purchasing pattern, it forecasts automatically, without any human intervention needed. No errors, no delays, nothing to worry about. Combining these advanced forecasts with convenient web dashboards, you will easily manage your stocks and save time and effort.

We are developing the NeuroCommerce platform for making decisions regarding the management of inventory. Unlike conventional formula-based approaches, our solution uses machine learning algorithms that are capable of analyzing past purchases and accurately predicting future ones based on this experience.

With each new purchase, the algorithm is capable of forecasting demand, checking the current state of the inventory and alerting the user when it is necessary to replenish supplies. In the end, we strive to develop a safe and affordable prototype for completely automated retail replenishment.

2. Literature Review / Background Study

A lot of researchers have already looked into using AI and machine learning for retail and demand forecasting. For example, Smith and Taylor [1] developed an algorithm that allowed making predictions on customer behavior to avoid running out of products in retail shops. They managed to prove that artificial intelligence can do the trick; however, it had problems with unorganized information.

Next, one may mention Johnson et al. [2], who focused their attention on deep learning and its use in e-commerce inventories. The method showed great results yet proved to be extremely complex to implement with a large demand for expensive computing resources. Then, there are Patel and Kumar [3], who introduced an automated stock tracking platform that utilizes simple regression models to alert administrators. Their system provided decent accuracy but required complex deployment processes and expensive third-party ERP integration.

In the past, the study by Brown [4] demonstrated the suitability of the Random Forests approach for addressing unpredictable market fluctuations. Lee managed to establish a solution to integrate light web technologies directly into the Python machine learning prediction algorithms, resulting in highly flexible and responsive systems. In essence, such integration became an industry standard to develop modern AI-based inventory management systems.

The problem is that most of the systems currently available in the market presuppose the establishment of an extensive enterprise environment, costly cloud subscription services, or both. That is why I have decided to focus on developing a more compact and budget-oriented prototype that will be capable of delivering the results of machine learning models locally.

3. Problem Statement

Inventory management is posing itself as one of the biggest problems for SMEs nowadays. The problem here is that most of the stores have not yet developed a method of estimating inventory through technology and have stuck with manual counting and recordkeeping. It's very inefficient, and at the same time, it can easily cause some miscountings and missing trends.

Thus, entrepreneurs are simply making guesses based on their intuition. This is exactly what I'm going to solve using the developed inventory management system powered by machine learning technologies and dashboard alerts that would help track demand.

4. Objectives

- To design and implement a machine learning-based retail automation prototype within four months.
- To develop an AI model (like Random Forest or XGBoost) that accurately predicts future product demand based on historical data.
- To create a web-based interface for inventory monitoring, data upload, and real-time visualization.
- To evaluate system functionality and predictive accuracy through rigorous testing scenarios.

5. Proposed Methodology / System Design

The proposed system follows a layered architecture consisting of a frontend, backend, database, and a machine learning engine.

System Flow:

1. Admin uploads daily sales data (CSV/Excel) or connects to a live database.
2. Backend reads the data and processes missing values or anomalies.
3. The Machine Learning Engine analyzes historical patterns to generate future demand forecasts.
4. Backend evaluates the predicted demand against the current stock in the database.
5. If stock is projected to fall below the optimal threshold, the system flags the item.
6. Automation module sends a low-stock alert to the dashboard.
7. For everyday verification, the admin views real-time graphs on the web portal.
8. The system continuously refines its model as new sales data is fed into the loop.
9. If data matches acceptable parameters, the forecast is validated; otherwise, it adjusts the confidence score.

6. Expected Outcomes

- A fully functional web-based inventory intelligence dashboard.
- A deployed and trained machine learning model capable of demand forecasting.
- Successful automated low-stock detection and alert demonstration.
- Technical documentation and system accuracy evaluation report.

7. Tools and Technologies to Be Used

- HTML, CSS, JavaScript (Bootstrap, Chart.js)
- Python
- Flask/ Django Framework
- Scikit-Learn (Machine Learning)
- Pandas & NumPy (Data Processing)
- PostgreSQL / MySQL
- Git & GitHub

8. References

- [1] J. Smith and R. Taylor, "Predictive Analytics in Retail: A Machine Learning Approach," Journal of Retail Tech, 2021.
- [2] A. Johnson et al., "Deep Learning Applications in E-Commerce Inventory," IEEE Transactions on AI, 2022
- [3] S. Patel and V. Kumar, "Automated Stock Tracking Platforms for SMEs," International Journal of Supply Chain Management, 2020
- [4] M. Brown, "Random Forest Models for Volatile Market Fluctuations," Data Science Quarterly, 2019.
- [5] C. Lee, "Integrating Lightweight Web APIs with Python Predictive Engines," Software Engineering Journal, 2023.

9. Work Division and Team Management

Member	Responsibility
Member 1	Machine learning model development and dataset processing
Member 2	Backend development, API routing, and AI integration
Member 3	Frontend web development and database management

Weekly team meetings will be conducted to monitor progress.

10. Timeline

Week	Activities
Week 1	Problem finalization, requirement analysis, literature review, project planning, and retail dataset identification.
Week 2	System architecture design, technology setup, database schema implementation, exploratory data analysis (EDA), and initial ML model selection.
Week 3	Core Machine Learning model training, backend API development (Flask/ FastAPI), frontend analytics dashboard creation, and UI/UX charting.
Week 4	System integration (connecting ML engine with frontend/backend), accuracy and functional testing, bug fixing, final report preparation, and presentation design.

11. Cost-Benefit Analysis

In regard to the technology stack used in this research paper, everything is going to be open-source, using machine learning algorithms that do not require any payment whatsoever. The website will be set up on my local machine, meaning no need to purchase any pricey enterprise software licenses or expensive cloud services to bring everything up and running. While one would indeed have to pay at some point to launch such a service into the wild, it won't cost anything in a college setting.

12. Constraints and Limitations

- Prototypes are limited to the local server environment and available for computing power.
- Scalability and model inference speed are not evaluated for massive, real-time institutional POS deployment.
- Requires clean, consistent historical data for the AI model to maintain accuracy.
- Security depends on proper admin authentication mechanisms.

13. Societal Impact

This system reduces much waste in the sense that stores will be prevented from ordering too much of something that will only be stored for an indefinite period and will eventually expire. The project was conceived to make things easier for small firms by providing managers with highly accurate automation. The overall idea behind the project is to make small shops compete with large retail giants.

14. Complex Engineering Problem Mapping

Table 1: Mapping complex problem -solving.

WP1 Dept. of Knowledge	WP2 Range of Conflicting Requirements	WP3 Depth of Analysis	WP4 Familiarity of Issues	WP5 Extent of Applicable Codes	WP6 Extent of Stakeholder Involvement	WP7 Interdependence
√	√	√	√	√	√	√

15. Complex Engineering Activity Mapping

Table 2: Mapping complex engineering activities.

EA1 Range of resources	EA2 Level of Interaction	EA3 Innovation	EA4 Consequences for society and environment	EA5 Familiarity
√	√	√	√	√

16. Green Computing Considerations

I designed the system to use optimized algorithms like Random Forest instead of superheavy deep learning neural networks. This cuts down a ton on CPU processing power and saves a lot of energy when training the model. Plus, since I'm just testing everything locally on my own computer, it avoids wasting server energy from constantly pinging the cloud.

17. Lifelong Learning and Personal Development

This project improves skills in artificial intelligence, backend framework design, data science, and full-stack web integration. This project also helps develop team spirit, problem-solving skills, and research-driven thinking, which will help me grow professionally in the future.